Interactive Sinhala Text Reader for the Visually Impaired with the assistance of OCR, NLP, and AI.

PP2 Comments and Actions Taken

Student ID	PP2 Panel Comments	Actions Taken
IT22072238 Muthunayaka M.J.J	Good Communication. Need to maintain the proper logbooks.	Done
IT22004468 Sathsara S.P.Y.B	Student doing Personalization. Personalization is only on the style of the answer presented. See if this can be improved.	Intent detection plus added delivery improvement with speeds and pauses (Adaptive TTS). Performs Routing and Turns policies to split request.
IT22259134 Madhusanka H.P.I	Need to maintain the proper logbooks.	Done
IT22120670 Devindi W.G.N	Need to maintain the proper logbooks. See if this can be made to a more practical component that is more usable.	Converted into an interactive RAG-based Q&A component. The reader can now ask questions and receive an answer, including follow-ups.

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The Problem and Solution

Problem

- People with Visual Impairment in Sri Lanka have limited access to information.



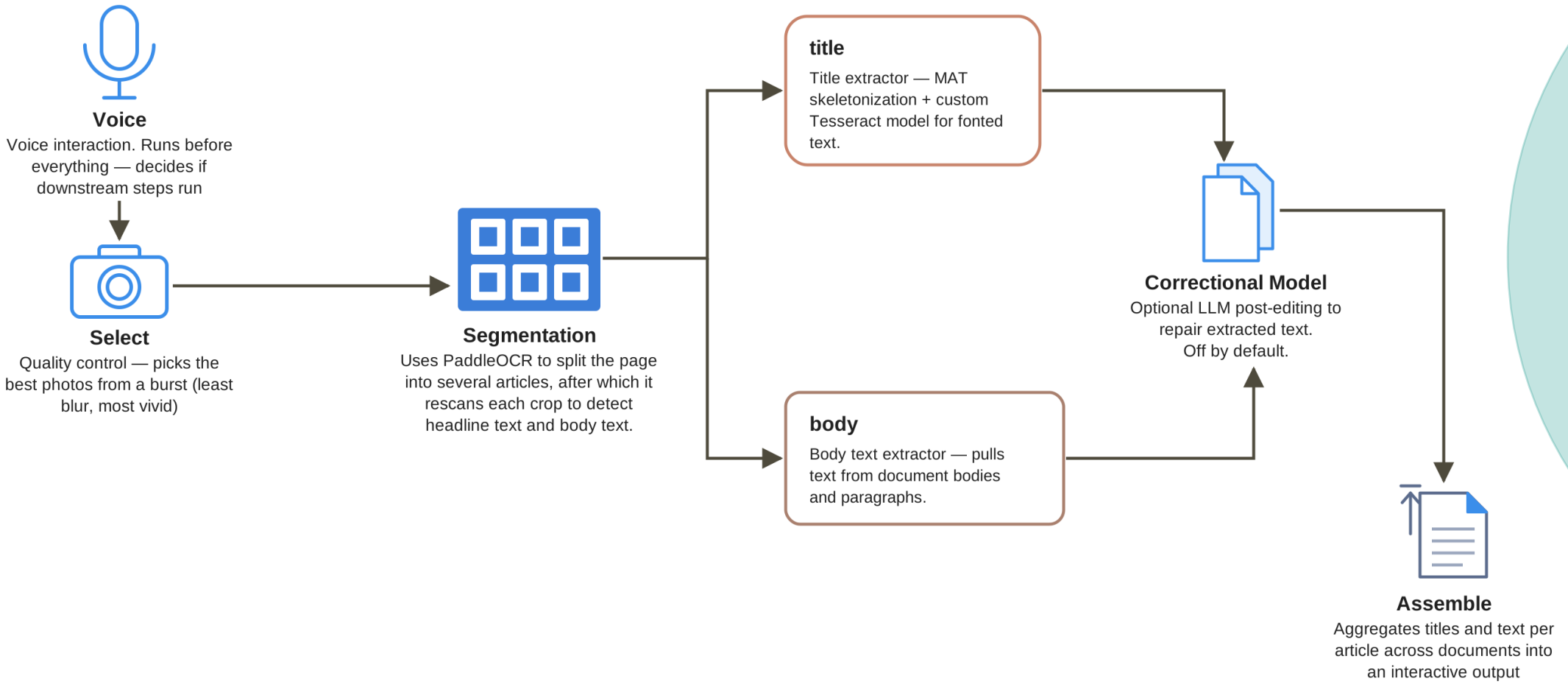
Statement

- Unlike existing assistive readers, which are mostly English-only, read text line by line, or need costly dedicated hardware, our solution runs on an ordinary phone. It works end to end in Sinhala: it understands newspaper layout, corrects OCR errors, answers spoken questions about the article and adapts to each user. The result is a hands-free reading assistant built for visually impaired Sinhala speakers.

Users

- Primary users are visually impaired and elderly Sinhala-speaking individuals who rely on newspapers, textbooks and other documents for daily information
- Secondary users - OCR developers and research institutions seeking a font-invariant Sinhala recognition, context aware Error correction, conversational Sinhala voice modules for integration.
- Tertiary User-Sri Lankan disability advocacy groups, government education agencies, and the open-source NLP community.

System Architecture Component 1 & Component 2





Topological Sinhala Character Recognition

Via Medial Axis Transform

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Brief Description

- Makes use Medial Axis Transform to skeletonize stylized Sinhala characters usually found in Title/Headline/Topics.
- Uses Paddle OCR to detect titles and crop with a margin, after which CRAFT detects characters through density and refines crop.
- Finally, it is binarized and skeletonize before fed to model.

Details

- Tesseract trained on ~90,000 images of the SinOCR dataset
- Images reseized to a height of 384 pixels, skeletonized to a 1 pixel, then dilated of 3 pixels.
- Binarize and stripping to a single sentence.

Results

- MAT trained performed better than a regular model on 29 of 135 fonts (21.5%).
- The 29 MAT-favored fonts had a 27.4% higher geometric mean edge density than regular-favored fonts, statistically significant at p = 0.0067
- In a 404-trial perceptual survey, images in which MAT model performed better was deemed by participants to be more complex(54.5%) establishing a suggestive trend (p = 0.0815)
- Out of the 8,872-images tested; 1584 independent victories,1977 Perfect Predictions, 1522 tied images

Context-Aware Sinhala Post OCR Error Correction



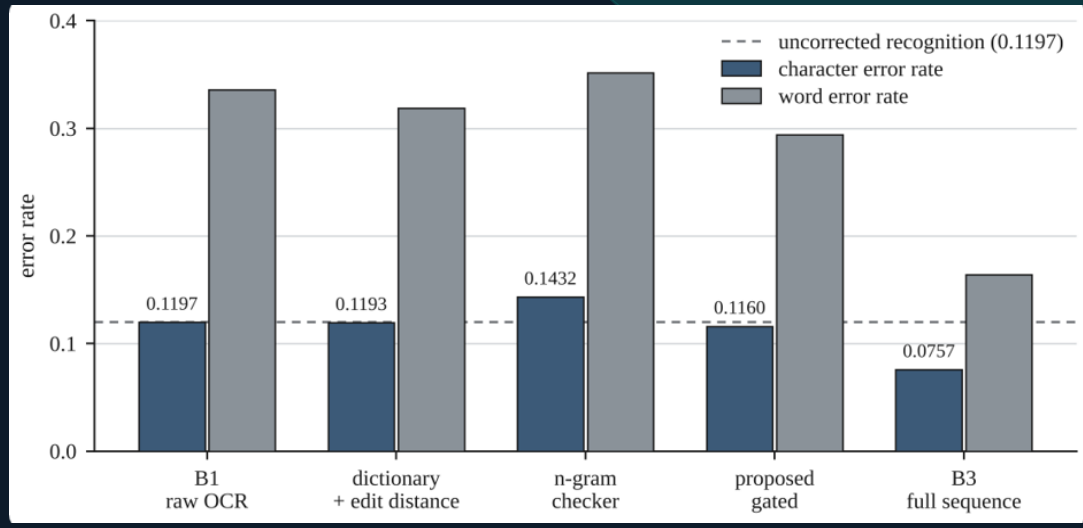
Brief Description

- Repairs Sinhala OCR errors that a dictionary cannot see. a damaged vowel sign usually produces another valid Sinhala word, so only the surrounding sentence reveals the error.
- Uses a fine-tuned mT5-small model that reads the whole sentence as context, rather than checking words one at a time.

Details

- Corpus: 1232 photographs of 3 Sinhala newspapers, all pages hand-transcribed.
- mT5-small fine-tuned on 36,743 synthetic OCR–truth pairs, generated from a 1192 entry confusion taxonomy measured on real newspaper photographs.

Results



Character error rate

0.1197 → 0.0757

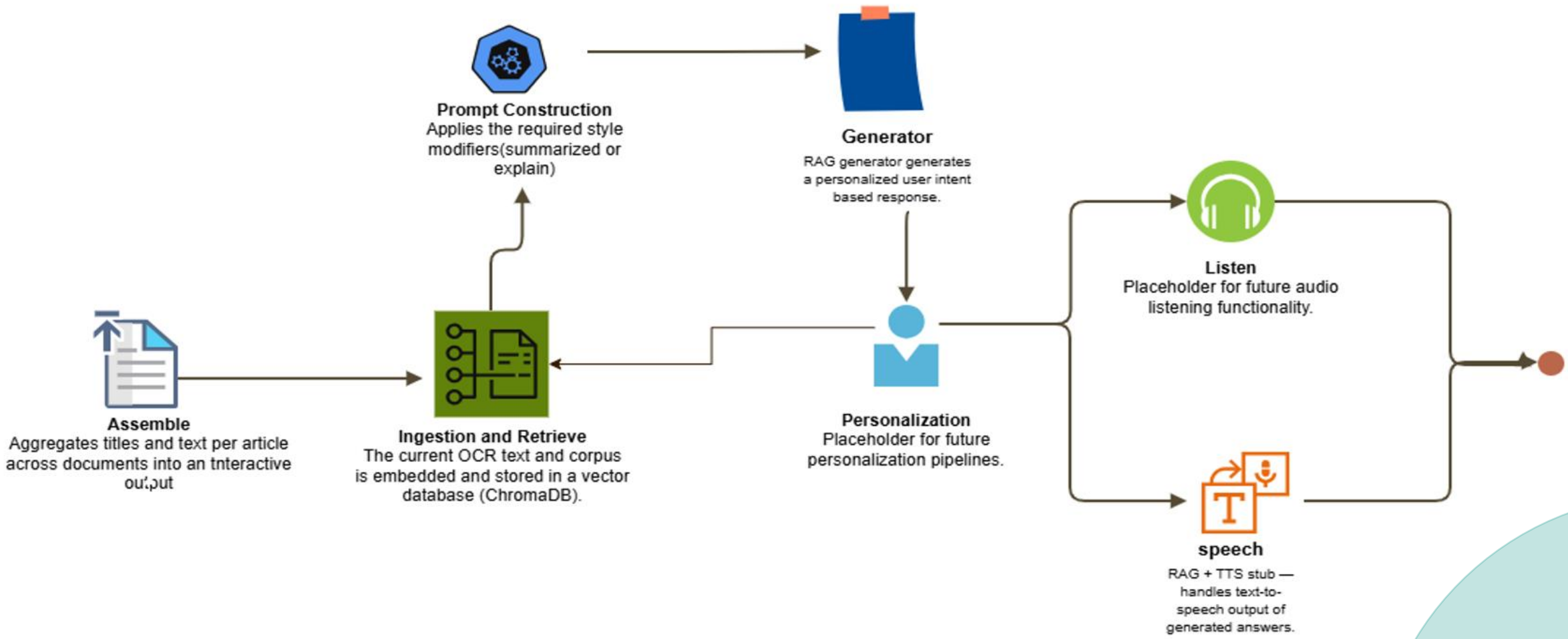
−36.7%

Word error rate

0.3358 → 0.1640

−51.2%

System Architecture Component 3 & Component 4





Advanced Document Retrieval and Information Generation

Brief Description

- Adaptive answer generation with use of Intents (Summarize, Elaborate) and styles(Simple, Detailed, Step by Step).
- Hybrid retrieval (Dense + BM25) over Sinhala articles and OCR text
- LLM generates the final answer in Sinhala from the retrieved evidence

Details

- Sinhala newspaper article corpus (headline, body, section category, publication date, source URL)
- **Inputs:** OCR corrected text and Voice output (route, intent, English query, style class, prompt modifier, detail level)
- `intfloat/multilingual-e5-small` embeddings stored in ChromaDB (dense search), BM25 over word + character 3/4-grams (keyword search for Sinhala inflections), and GPT-5.4-mini for answer generation.

Results

- Hybrid retrieval (Dense + BM25 + RRF) improved answer accuracy from 51.8% (dense only) to 73.2% on 72 answerable questions.
- Hybrid retrieval retrieved the correct chunk within the top 5 results for 92.9% of questions, compared to 58.9% for dense retrieval.

Adaptive Conversational Personalization and Voice Interaction Module

(Personalization and Voice Interaction Module)



Brief Description

- Adapts each answer's style (Simple / Detailed / Step-by-step) and voice (speed, pauses) to each user.
- NLLB-600M translates Sinhala to English; Llama 3.2 1B detects the intent.
- A per-user River online-learning model learns each user's style from their own requests, with no pre-trained dataset. The result is sent to the RAG as a prompt instruction or handled locally (replay at a new speed).

Details

- River SoftMax Regression, one model per user
- Learns only from explicit requests
- History decides until 3 signals
- Slower / faster: speed ×0.90 / ×1.10
- Repeat: speed ×0.95 + more pauses
- Speed kept within 0.8–1.3

Results:-

Approach	Accuracy	Avg Latency
A1 - NLLB + Llama 3.2:1b	**77.8%**	1.53 s
A2 - Qwen 2.5:1.5b (direct)	2.2%	5.20 s
A3 - Trained (TF-IDF + LinearSVC)	62.2%	0.93 s

User Voice Input	Route	Evidence Source	User Profile History	Results
"Explain this document?"	GENERATE by RAG	Default Style	Simple: 0.0 | Detailed: 0.0 | Step: 0.0 Speed 1.00 · Pause 1.0	Detailed (Default) Cold start, nothing learned yet
"Explain this document in detail?"	GENERATE by RAG	Style word "detail"	Simple: 0.0 | Detailed: 1.0 | Step: 0.0 Speed 1.00 · Pause 1.0	Detailed (Model Learns); learned
"Read it slowly?"	ACTION Local, no RAG	Speed word "slowly" (delivery, not style)	Style unchanged Speed 1.00 → 0.90 Pause 1.0 → 1.1	Saved answer replayed slower (replay at 0.90×)
"Repeat it again?"	ACTION Local, no RAG	"Repeat" (didn't hear)	Style unchanged Speed 0.90 → 0.855 Pause 1.1 → 1.2	Saved answer replayed again

User Acceptance

- **Proxy Testing:** System usability was validated via developer-simulated visual impairment (eyes-closed navigation & audio prompts).
- **Demonstration Success:** The full, integrated system was successfully demonstrated and recorded for future purposes.
- **Full Implementation:** While it covers the requirements in principle it is yet to be tested with blind individuals

Benchmark Tests

Integration Stability: 439 of 449 automated tests passed successfully (5 failing, 5 skipped for missing capture data).

End-to-End Latency and Task Success:

-The full reading path (shutter to first speech) takes ~113 seconds at the median on CPU and ~39 seconds at the median GPU (6gb).

-Local voice commands (read, next, repeat) take 3–5 seconds, and generated answers (summarize, question, simpler) take 26–36 seconds.

-287 of 288 scripted voice turns (99.6%) took the correct action, and 36 of 41 real phone captures (88%) produced a readable document.

Accuracy Gap: While there is no method to full check the accuracy of the system, each component individually can

-In image text extraction, **Component 1:** 94.1%-(CER 0.0586) and **Component 2:** 83.6%-(WER 0.1640) and 92.4%-(CER 0.0757).

-In generation and interaction, **Component 3:** 90.2% answer faithfulness and **Component 4:** 85.4% intent accuracy.

Limitations of the System

- **Architecture Fragmentation:** Separate processes because components have different incompatible Python environments.
- **Error Propagation:** Uncorrected OCR errors are passed downstream, which can negatively impact the accuracy of retrieval & generation
- **Consistency:** Some articles work well while others struggle

Future Improvements

- **Optimization** : Focusing on improving the performance and latency of the whole application and reducing the waiting periods.
- **Target Audience Validation**: Execute the planned formal studies with blind and low-vision Sinhala readers to measure real-world comprehension and satisfaction
- **Complete system testing:** Run an instrumented end-to-end accuracy test with ground-truth assistance to measure compound error rates, with true deployment in place.

Thank You!

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